Session 01: AI Engineering Landscape

One-hour teaching pack for UK working professionals with basic Python knowledge

Why this session matters. Many organisations start with a tool or model and only later ask whether the data, workflow, evaluation, deployment, and governance are ready. This session gives learners a practical map for deciding what kind of AI system a workplace task needs before code is written.

60-minute rhythm

Time	Activity	Purpose
0-10 min	Welcome, course orientation, needs assessmt	Connect AI engineering to learners' own work contexts.
10-30 min	Theory: AI, ML, DL, GenAI and the AI engine	Build a shared vocabulary and decision framework.
30-50 min	Guided practical: map a business process int	Practise thinking like an AI engineer using Python, notebooks and GitHub
50-60 min	Discussion, quiz, assignment briefing	Check understanding and set up self-directed homework.

Problem → **Data** → **Model** → **Interface** → **Evaluation** → **Deployment**

AI engineering is an operating workflow, not just a model-training activity

Course alignment

This session begins the 30-session programme objective: learners will progressively learn to design, build, evaluate, deploy, and monitor modern AI systems. The timetable positions Session 01 in Module 1 as the landscape session covering AI vs ML vs DL vs GenAI, what an AI engineer builds, system layers, and UK workplace use cases.

Instructor note: no previous AI knowledge is assumed, but learners should already be comfortable opening notebooks, reading simple Python, and saving files to GitHub.

1. Learning objectives and needs assessment

By the end of the session, learners should be able to:

- Explain the AI engineering workflow from problem framing through deployment and monitoring.
- Distinguish AI, ML, deep learning, and generative AI using workplace examples.
- Identify the data, model, product, evaluation, deployment, and operations components of an AI use case.
- Decide whether a task is better suited to predictive ML, GenAI, both, or neither.
- Create a one-page AI use-case canvas suitable for discussion with a manager, data owner, or technical team.

Quick needs assessment (5 minutes)

Prompt	Learner action	Instructor listens for
Where in your work do people make repeated	Write one process on a sticky note	Candidate predictive ML tasks.
Where do people spend time reading, writing, summarising, or extracting information?	Write one text-heavy process.	Candidate GenAI tasks.
What would make AI unsafe or unacceptable	Name one risk.text?	Privacy, bias, human oversight, accountability.

Recap of assumed prior knowledge

Learners already know basic Python syntax, functions, notebooks or IDE usage, simple data structures, and basic command-line use. The session therefore avoids Python refreshers. Instead, the notebook uses familiar operations reading a CSV with Pandas, filtering rows, creating a dictionary, and saving a Markdown file - to build AI engineering judgement.

Adult-learning design choices

The session starts from real workplace processes rather than abstract algorithms. Learners use their own domain expertise to judge whether a proposed AI system would be useful, measurable, and governable. The practical task is deliberately lightweight in code but demanding in thinking, because good AI engineering begins with problem definition, not model selection.

2. AI, ML, deep learning and generative AI

A useful definition of an AI system is a machine-based system that infers from inputs how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments. In class, translate this into: an AI system takes inputs, performs learned or rule-supported inference, and produces outputs that affect a workflow.

Term	Plain-English meaning	Typical workplace output	Example
AI	Broad field of systems that perform tasks associated with human intelligence	Prediction, classification, automation	A tool that prioritises support tickets
Machine Learning (ML)	Systems learn patterns from historical data instead of being hand-coded for every rule	Score, class, forecast, ranking	Predicting customer churn
Deep Learning (DL)	ML using neural networks with many layers; strong for images, speech, text, and high-dimensional data	Image label, text embedding, detection, sequence output	Image recognition system
Generative AI	AI that generates new content or structured outputs from prompts and context	Summary, draft, extraction, answer, code, image, plan	Chatbot generating responses

Key distinction: predictive ML vs GenAI

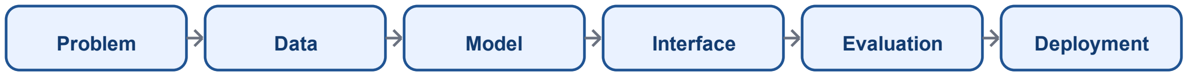
Predictive ML usually answers: 'What is likely to happen?' or 'Which category does this belong to?' It is evaluated with metrics such as accuracy, precision, recall, MAE, RMSE, or calibration. **GenAI** usually answers: 'What useful content or structured response can be produced from this context?' It is evaluated with rubrics, factuality checks, human review, safety checks, and task-specific test sets.

Common mistake to prevent

Do not label every AI idea as 'a chatbot'. Some valuable AI systems have no chat interface at all: they may be batch forecasts, risk scores, dashboards, triage queues, document extractors, or API services embedded in an existing tool.

3. What an AI engineer builds

An AI engineer builds the bridge between data science, software engineering, product design, and operational governance. The job is not simply to choose a model. It is to turn a business process into a reliable system that produces useful outputs, can be evaluated, and can be operated safely over time.



AI engineering is an operating workflow, not just a model-training activity.

Deliverable	What it contains	Why it matters
Problem statement	User, decision, input, output, success measure, constraints	Defines scope and ensures clarity before building
Data specification	Sources, fields, labels, access, quality checks, privacy status	Ensures reliable, compliant, and usable data
Model or prompt design	Predictive model, LLM prompt, retrieval, rules, or hybrid design	Determines how effectively the system performs
Evaluation plan	Metrics, test cases, human review process, risk thresholds	Validates performance and manages risk
Interface design	Notebook, dashboard, API, form helper, chatbot, report, or workflow integration	Makes the system usable for end users
Operational plan	Versioning, monitoring, incident response, cost tracking, ownership	Keeps the system useful and safe after launch

Core delivables

Prevents building a technically impressive solution to the wrong problem. AI outputs are limited by data availability and governance. Converts inputs into outputs, with explicit assumptions. Defines what 'good enough' means before deployment. Ensures the output reaches users where work happens.

Vocabulary anchor

Model is only one component. An AI application also needs data flows, a human or software interface, evaluation evidence, deployment mechanics, and a plan for ongoing operations.

4. The four engineering layers

Data layer	sources, labels, quality, privacy, access, lineage
Model layer	predictive model, LLM, rules, prompts, evaluation
Product layer	user interface, workflow fit, feedback, adoption
Operations layer	monitoring, drift, incident response, cost, compliance

Data layer

The data layer includes data sources, labels, access permissions, lineage, quality checks, and data protection constraints. In a predictive ML use case, labels may be historical outcomes such as 'renewed' or 'did not renew'. In a GenAI use case, data may be a document corpus, policy library, CRM notes, or approved templates used as context.

Model layer

The model layer contains the predictive model, LLM, prompt, retrieval design, feature engineering, thresholds, and evaluation scripts. It should be versioned and reproducible. For this first session, learners do not train a model; they decide which model family would plausibly fit a process.

Product layer

The product layer turns model output into usable workplace behaviour. A high-performing model can fail if the output appears too late, lacks context, interrupts users, or is not trusted. Learners should ask: who acts on the output, where do they see it, and what decision changes as a result?

Operations layer

The operations layer covers monitoring, cost, access control, retraining or prompt updates, incident response, and accountability. It is where prototypes become services. It also includes version control in GitHub, repeatable notebooks, and clear ownership.

5. UK workplace use cases and approach selection

Use the question below to choose a first AI approach. The answer may be 'no AI yet' if the data is weak, the process is not measurable, or the risk is too high for an early prototype.

Workplace task shape	Better first approach	Reason	Example
Forecasting, scoring, ranking, classification	Historical examples can be used to train and test a model	Historical examples can be used to train and test a model	Credit risk scoring, demand forecasting
Summarising, drafting, extracting, rewriting	GenAI	The output is language or structured content derived from context	Email drafting, report summarisation
Score plus explanation or drafted action	Hybrid	A predictive model identifies priority; GenAI helps communicate or summarise	Lead scoring + personalised outreach email
One-off judgement, low volume, unclear outcome	No AI yet	Automation may add complexity without measurable value	Rare strategic decisions

Examples for discussion

- **NHS operations:** predictive model for missed appointment risk; GenAI for approved, human-reviewed patient communication drafts.
- **Local government:** text extraction from maintenance requests; predictive model for expected resolution time.

- **Finance:** spend forecasting with predictive ML; GenAI for narrative variance commentary.

- **Legal operations:** retrieval-based GenAI for contract summaries with citations, not free-form legal advice.

- **Customer service:** priority classification plus complaint summary, with human approval before sending.

Discussion prompt

Pick one process from your role. Is its main output a prediction, a generated text, a structured extraction, or a workflow action? What evidence would convince your manager that it works?

6. Governance, ethics and responsible delivery

Responsible AI is not a separate topic bolted on at the end. It affects the first problem framing decision: what data may be used, what outputs are acceptable, who remains accountable, and how harms are detected.

Governance question	Why it matters	Session 01 learner action
Is personal data involved?	UK GDPR and data protection obligations may apply, especially for employees, customers, patients, or tenants.	Mark the data source and note the owner.
Could the output affect rights, access, money, work, or safety?	Higher impact use cases need stronger evaluation, explanation, oversight, and escalation.	Identify who is affected and who ca override the system.
What is the failure mode?	Predictive ML can misclassify; GenAI can hallucinate, omit context, or produce unsafe content.	Write one likely failure and one mitigation.
How will users know when not to trust it?	Adoption requires calibrated confidence, source display, and clear limitations.	Define a human review step for early prototypes.
What evidence is retained?	Audits require versions, prompts, data snapshots, decisions, and evaluation results.	Record what evidence will be stored and how it will be accessed.

Use GitHub and notebooks to preserve reproducible artefacts.

Practical governance anchor

For GenAI, ask for source-grounding and human review before workplace action. For predictive ML, ask whether the model is calibrated, whether errors are unevenly distributed across groups, and whether the threshold aligns with business risk. For both, avoid using data in ways people would not reasonably expect.

Sources used in this pack

OECD AI Principles definition of AI system; NIST AI RMF Generative AI Profile; UK ICO guidance on AI and data protection; UK AI Opportunities Action Plan. Full links are listed in the final page references.

7. Practical walkthrough: map a process in a notebook

The practical activity uses Python, a Jupyter notebook, a cleaned CSV dataset, and GitHub-style project structure. The goal is not to train a model. The goal is to make the first engineering judgement: how a business process decomposes into data, model, interface, evaluation, and deployment components.

Step 1 - open the project

- Open the folder **Session_01_AI_Engineering_Landscape/**.

- Launch the notebook **Session_01_AI_Engineering_Landscape.ipynb**.

- Confirm the **data/** folder contains the example CSV and blank canvas.

- Create a GitHub repository or local folder that preserves the same structure.

Step 2 - load the process examples

```
from pathlib import Path
import pandas as pd

DATA_PATH =
Path("data/business_process_ai_mapping_examples.csv") examples =
pd.read_csv(DATA_PATH) examples[["process_id",
"business_process",           "predictive_ml_candidate",
"genai_candidate"]].head()
```

Step 3 - interpret the columns

Column	Interpretation
business_process	The workflow or decision area being considered.
data_inputs	Evidence available to the AI system.
predictive_ml_candidate	Possible score, label, forecast, or ranking task.
genai_candidate	Possible drafting, summarisation, extraction, or transformation task.
interface	Where the output reaches a colleague or system.
evaluation_metric	How the class would judge quality.
deployment_route	How the idea could move beyond a notebook.

governance_risks	Ethical, data protection, fairness, safety, or accountability concerns.

Mini-project: build an AI component map

Learners now choose one example process or one process from their own workplace. They fill in a component map that can later become a README, an experiment plan, or an initial product discovery document.

Instructor demonstration

```
selected = examples.loc[examples["process_id"] == "P002"].iloc[0]

component_map = {     "data":
selected["data_inputs"],
    "model": {
        "predictive_ml": selected["predictive_ml_candidate"],
        "genai": selected["genai_candidate"],
    },
    "interface": selected["interface"],
    "evaluation": selected["evaluation_metric"],
    "deployment": selected["deployment_route"],
    "governance":
selected["governance_risks"], } component_map
```

Learner exercise

```
my_canvas = {     "Business process": "Replace with your
workplace process",
    "Data inputs": "List existing data sources and owners",
    "Model approach": "Predictive ML, GenAI, both, or neither",
    "Interface": "Dashboard, API, form helper, chatbot, notebook, etc.",
    "Evaluation": "Metric, review rubric, business KPI, or risk
threshold",
    "Deployment": "Notebook prototype, batch job, API, internal app,
etc.",     "Governance": "Privacy, bias, safety, accountability, human
review", }

pd.DataFrame(my_canvas.items(), columns=["Field", "Your notes"])
```

Pair discussion prompts

- Which component was easiest to define: data, model, interface, evaluation, deployment, or governance?

- Where did you make an assumption that would need confirmation from a colleague?

- Would your use case be predictive ML, GenAI, both, or neither? Defend your answer in one sentence.

- What is the smallest safe prototype you could build next?

Expected practical output

A completed AI use-case canvas saved as Markdown or CSV, plus a notebook cell showing the component map. Learners should be able to explain every field in plain English.

8. Discussion, assignment and next step

Question	Strong answer should mention
What is the difference between predictive ML and G	GenAI?
Why is the interface part of AI engineering?	The output only creates value when it changes a real workflow for a real user.
Why is evaluation defined before deployment?	It prevents subjective success claims and clarifies risk thresholds.
What does GitHub add at this stage?	Versioning, reproducibility, collaboration, README, and audit trail.

Knowledge check (5 minutes)

Prediction or classification from historical examples vs generation or extraction of content from prompts/context

Homework: one-page AI use-case canvas

- **Deliverable:** a one-page canvas for a workplace process you understand.
- **Required fields:** process, data inputs, model approach, interface, evaluation, deployment, governance risks.
- **Evidence:** include at least one reason why your use case is predictive ML, GenAI, both, or neither.
- **Submission:** commit the canvas and notebook to GitHub, or export as PDF/Markdown if GitHub is unavailable.

Evaluation criteria

Criterion	Meets expectations
Clarity	A non-technical colleague can understand the process and intended AI output.
Approach fit	The learner correctly distinguishes predictive ML from GenAI.
Engineering completeness	All system components are represented, not just the model.
Governance awareness	At least one realistic risk and mitigation are included.
Reproducibility	Notebook, data, and output are saved in a clear folder structure.

References

Course timetable: AI Engineering for Working Professionals, Session 01. OECD AI Principles, AI system definition: https://oecd.ai/en/ai-principles. NIST AI RMF Generative AI Profile, 2024: https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence. UK ICO Guidance on AI and data protection: https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/artificial-intelligence/guidance-on-ai-and-data-protection/. UK AI Opportunities Action Plan: https://www.gov.uk/government/publications/ai-opportunities-action-plan/ai-opportunities-action-plan.

Next session link

Session 02 will convert this landscape thinking into data collection and problem framing: labels, targets, data quality, leakage, ethics of collection, and train/validation/test thinking.